TCE 318P — Mathematics PK–5: Child & Curriculum Integration

Miami University · Fall 2026

Instructor and logistics

Instructor	Dr. Michael Todd Edwards
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Office	401H McGuffey Hall, Oxford campus
Office hours	Monday, Tuesday, Thursday and Friday, 1:00–3:00 p.m., by appointment. Google Meet by default; the link comes with the booking confirmation. In-person alternative: Rentschler Library, Hamilton campus, Tuesdays and Thursdays; 401H McGuffey Hall, Oxford campus, Mondays and Fridays.
Sections	**H A** (CRN 12886) · Hamilton campus **A** (CRN 12918) · Oxford campus
Class meets	**Hamilton (H A)** — Tuesdays & Thursdays, 11:40 a.m.–1:00 p.m., 216 Mosler Hall **Oxford (A)** — Mondays & Wednesdays, 10:05–11:25 a.m., 126 McGuffey Hall
Term	August 24 – December 11, 2026
Canvas	Oxford: https://miamioh.instructure.com/courses/258372 Hamilton: https://miamioh.instructure.com/courses/258364
Course home	The public course site carries the schedule, daily outlines, deliverables, policies, and tools. Work is submitted and grades are posted in Canvas.

Term detail. Hamilton first meets Tuesday, August 25 and last on Thursday, December 3. Oxford first meets Monday, August 24 and last on Wednesday, December 2. Monday, September 7 is Labor Day (no classes; campuses closed) — Oxford's Week 3 meets Wednesday only; Hamilton is unaffected. Finals week is December 7–11; the Final Report and Review & Reflection are due Wednesday, December 9.

Course description

TCE 318P is a methods course for preservice PK–5 teachers, focused on teaching mathematics for understanding. Candidates design, enact, and revise lessons that are responsive to students' developmental needs, mathematical thinking, and lived experiences — learning to read curriculum materials, assess student work, and reflect on practice through field-based experience.

The arc of the semester. The whole course builds toward one act of evidence-based inquiry: candidates develop a question about how students understand a mathematical concept, build a lesson to investigate it, take it into a real classroom, and report honestly on what they found.

- **Act I — Foundations (Weeks 1–5).** The core moves: reasoning, listening and questioning, geometry and van Hiele, differentiation, and assessment.
- **Act II — Build the investigation (Weeks 6–10).** A concept, a research question, and a nested Before–During–After lesson designed to reveal student thinking.
- **Act III — Field and report (Weeks 11–15).** Enact the lesson, collect student work, and communicate what was learned in a final report.

Course objectives

By the end of this course, candidates will be able to:

- Design and adapt lesson plans that align with standards, address developmental progressions, and respond to student thinking.
- Implement effective teaching practices, including questioning, active listening, and facilitation of rich mathematical tasks.
- Analyze and assess student work to inform instructional decisions.
- Use observation and field experience to plan targeted instruction.
- Integrate principles of access, inclusion, and differentiation into mathematics instruction.
- Engage in reflective practice to improve teaching and communicate professional insights.

Course materials

There is no textbook to purchase. Readings are short NCTM practitioner articles accessed through JSTOR via Miami's library, listed in the schedule below.

Course tools are browser-based; there is nothing to install and no account to create. Any current browser works (Chrome, Edge, Firefox, Safari) on macOS, Windows, Linux, or a Chromebook. A laptop or desktop is recommended, since the Reading Journal is turned in by printing to PDF from the browser.

The apps used across the semester are Journaler (reading and weekly Reading Journal), Research Question Studio (Week 7), the Metacognition suite — Metadoku, Metaplete, Metagons — Pentomino Builder (Week 4), and nVoke and Sort Lab (Week 3).

Schedule

The course runs August 24 – December 11, 2026. Hamilton meets Tuesdays and Thursdays through December 3; Oxford meets Mondays and Wednesdays through December 2, and not on Labor Day, Monday September 7.

Act I — Foundations

Week	Dates	Focus	Reading
1	Hamilton Aug 25, 27 · Oxford Aug 24, 26	Reasoning puzzles	McFeetors & Palfy, "Playing Games in Math Class"
2	Hamilton Sep 1, 3 · Oxford Aug 31, Sep 2	Listening & questioning	Reinhart, "Never Say Anything a Kid Can Say!"; Switzer, "Hundred Chart Challenge"
3	Hamilton Sep 8, 10 · Oxford Sep 9	Geometry & van Hiele	Newell & Orton, "Classroom Routines"; Higgins, "Building a Team of Learners"
4	Hamilton Sep 15, 17 · Oxford Sep 14, 16	Differentiation & measurement	Edwards, Meagher & Özgün-Koca, "Pentominoes"; Simpson et al., "Rethinking Teachers' Questioning"
5	Hamilton Sep 22, 24 · Oxford Sep	Assessment & rubrics	McGatha & Darcy, "Rubrics at Play"

Week	Dates	Focus	Reading
	21, 23		

Act II — Build the investigation

Week	Dates	Focus	Reading
6	Hamilton Sep 29, Oct 1 · Oxford Sep 28, 30	Field dip & the concept profile (in the field)	Reading Journal due — all five entries
7	Hamilton Oct 6, 8 · Oxford Oct 5, 7	Research questions	—
8	Hamilton Oct 13, 15 · Oxford Oct 12, 14	Rich tasks	—
9	Hamilton Oct 20, 22 · Oxford Oct 19, 21	Math routines	—
10	Hamilton Oct 27, 29 · Oxford Oct 26, 28	The investigation plan	—

Act III — Field and report

Week	Dates	Focus	Reading
11	Hamilton Nov 3, 5 · Oxford Nov 2, 4	Field enactment (in the field)	—
12	Hamilton Nov 10, 12 · Oxford Nov 9, 11	Field enactment (in the field — no seminar)	—
13	Hamilton Nov 17, 19 · Oxford Nov 16, 18	The week back: share-out & writing	Research candidates gather themselves, for “What does research say?”
14	Nov 23–26	No class — Thanksgiving	—
15	Hamilton Dec 1, 3 · Oxford Nov	Review, potluck & share	Candidates’ own sources

Week	Dates	Focus	Reading
	30, Dec 2		
Finals	Wed, Dec 9	Final Report and Review & Reflection due, via Canvas	—

Deliverables and grading

The course is worth **350 points**. Everything builds toward the field investigation and its report.

Deliverable	Hamilton (T/Th)	Oxford (M/W)	Points
Act I — Foundations			
Pedagogical Excursion #1	Sun Aug 30	Sun Aug 30	20
Pedagogical Excursion #2	Thu Sep 3	Wed Sep 2	20
Pedagogical Excursion #3	Sun Sep 13	Sun Sep 13	20
Pedagogical Excursion #4	Sun Sep 20	Sun Sep 20	20
Pedagogical Excursion #5	Thu Sep 24	Wed Sep 23	20
Reading Journal	Sun Sep 27	Sun Sep 27	50
Act II — Build the investigation			
Concept Profile	Sun Oct 4	Sun Oct 4	20
Research Question	Sun Oct 11	Sun Oct 11	25
Rich Task	Sun Oct 18	Sun Oct 18	25
Math Routine	Sun Oct 25	Sun Oct 25	25
Investigation Plan	Sun Nov 1	Sun Nov 1	25
Act III — Field and report			
Final Report	Wed Dec 9	Wed Dec 9	50
Review & Reflection	Wed Dec 9	Wed Dec 9	30
Total			**350**

Candidates are graded on evidence-based reasoning and honesty about limits — not on whether they “answered” their question. An honest “here is what I saw, and here is what I still don’t know” is strong work.

Grading scale

The grade is the total of the deliverables above, out of 350 points. A candidate earns the grade in the highest row whose point figure they have reached. Every possible total lands in exactly one row, including totals with a fraction: 304.5 reaches 294 but not 305, so it is a B.

Grade	Points (at least)	Percent	Grade	Points (at least)	Percent
A+	343	98%	C+	270	77%
A	329	94%	C	259	74%
A−	315	90%	C−	245	70%
B+	305	87%	D+	235	67%
B	294	84%	D	224	64%
B−	280	80%	D−	210	60%
			F	below 210	below 60%

Working with AI

On their own work, generative AI is the candidate's choice, not a requirement. When used, it is a thinking partner, not an answer generator — the reasoning has to be the candidate's.

- **Gemini is recommended**, since candidates have Miami Pro accounts. Any tool is fine; none is required.
- **Human first.** A person reads the work before a machine does. This is practiced all semester: candidates test their own thinking, and a colleague reads the draft, before an AI ever weighs in.
- **Disclose it.** Candidates state how they used AI in their work (a sentence is enough), and never present AI-invented facts or citations as real.
- **Todd-in-a-Can** is the course AI reviewer, built for the final paper. Candidates may run it as often as it helps; a human peer reads the draft as well.

Weeks 7–10 are the exception: there, one AI tool is part of the assignment. Those four weeks use the genAI planning partner built into **BDA Planner**, and each assignment says which part of the plan to run it on.

- **What it is.** A set of critics configured against each assignment's rubric — one for the research question, one for the rich task, one for the routine, one

for the whole investigation plan. Candidates receive an invitation before Week 7; there is nothing to buy.

- **A critic, not a designer.** It will not write the question, task, routine, or lesson, and it is built to refuse if asked. The design judgment is what is being assessed.
- **Human first, again.** Colleagues test the work *before* the genAI planning partner sees it. The "two reads" criterion on each rubric asks what the human read surfaced, then what the machine added, and what the candidate rejected from both.
- **Not a stand-in for children.** Any strategy it names is a possibility to watch for, never a fact about how a real child thinks.
- **Pseudonyms only.** Everything the genAI planning partner reads is de-identified first. No student names, schools, or identifying details — ever, in any tool.

Policies

Accessibility and Student Disability Services

If you anticipate or experience barriers based on disability in relation to the course design, you are encouraged to connect with Student Disability Services (SDS) and request reasonable accommodation. SDS provides accommodations and services for students with a variety of disabilities, including chronic illnesses, psychological, medical, learning, physical, and sensory disabilities, amongst others. I invite you to speak with me regarding your access needs while contacting SDS to learn more about the affiliation process and procedures for requesting accommodations.

- Oxford Campus: SDS@MiamiOH.edu
- Hamilton Campus: RADS@MiamiOH.edu
- Middletown Campus: RADS@MiamiOH.edu

Current SDS affiliated students should request accommodations according to SDS procedure. You are strongly encouraged to request and discuss your accommodation needs during the **first 1–2 weeks of the semester**.

Miami University's recommended statement — Student Disability Services syllabus statements, https://miamioh.edu/academic-programs/resources-services/student-disability-services/faculty-tools/syllabus-statements.html

Title IX, and what I am required to report

Reflective writing and conversations in this course sometimes touch on personal experience. There is one limit on confidentiality that I cannot set aside, and you should know about it in advance rather than afterwards.

I am a mandatory reporter. If you write or speak about having experienced sexual violence — including rape, sexual assault, dating violence, domestic violence, or stalking — and indicate that it happened while you were a Miami student, Miami requires me as a faculty member to report it to the University's Title IX office. I cannot promise confidentiality, and I must pass on relevant details, including names, if I know them. Someone from the University will then contact you about accommodations, support services, and your options for holding the person who harmed you accountable. You are not obliged to respond to them or to take any further step.

If you would rather that not happen, talk to a confidential resource instead of to me. These people can hear what happened without starting a report:

- **Women Helping Women** — 24-hour crisis line **513-381-5610** (call or text), toll-free **877-889-5610**. Campus advocate: **513-818-4894**, frieseo@MiamiOH.edu. Miami states that speaking with the campus advocate is private and **will not initiate a report** with the University.
- **Counseling — use your own campus.** Confidential and free of charge.

Hamilton	Oxford
Regionals Student Counseling Services — **513-727-3431**, Rentschler Hall.	**Student Counseling Service** — **513-529-4634**, StudentCounseling@MiamiOH.edu, weekdays 8 a.m.–5 p.m.
H.O.P.E. Line — **855-249-5649**, staffed 24 hours by a crisis support specialist. This is the after-hours number.	After hours, ask the university police dispatcher at **513-529-2222** for the on-call counselor.

If you do want to make a report, or to ask about University policy:

- **Title IX & Title VI Coordinator** — Danica Myers, Warfield Hall 017, **513-529-1424**, TitleIX@MiamiOH.edu. Reports may be made at any time, including outside business hours.
- **Office of Community Standards** — **513-529-1417** · **Dean of Students**, 110 Warfield Hall — **513-529-1877**

- **Campus security — use your own campus.** In an emergency, **911**.

Hamilton	Oxford
Allied Universal, **Mosler Hall 110**, **513-785-3222**.	**Miami University Police**, **513-529-2222** (24 hours).

What Title IX is. Title IX makes violence, harassment, and discrimination based on sex and gender civil-rights offenses, subject to the same accountability and the same support as offenses against other protected categories such as race or national origin. Miami University prohibits sex discrimination and sexual misconduct, including harassment, domestic and dating violence, sexual assault, and stalking.

Contacts verified against Miami's Title IX statement, Regionals Student Counseling Services, and Regional Campus Safety, July 2026. This course runs on two campuses, so the counseling and security numbers above are split by campus. Use the column for the campus you attend.

Academic integrity

Your work should reflect honesty and respect for learning. Cheating, plagiarism, fabrication, unauthorized collaboration, or misrepresentation are prohibited and handled under Miami University's Academic Integrity Policy. Cite all sources you use, **including AI assistance**.

Attendance

This is a Block course; participation in each element of the block is contingent on the others. Missing more than one seminar or more than three field-placement days — made up or not — can result in a Statement of Concern and may result in being dropped from all courses in the block.

Three required **Block 2 seminars** meet via Zoom (Field Placement Expectations · Block 3 overview · Student Teaching). Dates and the join link will be announced in class.

Late work

Anything may be up to 48 hours late. No penalty, no need to ask, and no explanation required. Submit it and move on.

If you need longer than that, email me. You will get it. I would rather read your real work a week late than a panicked version on time. There is nothing to prove and no documentation to produce.

Three things cannot slide, because someone other than you depends on them:

- **Drafts that go to peer feedback or a lesson-planning session.** A colleague has to read it, so it has to exist.
- **The Investigation Plan, Week 10.** You cannot walk into a room of real children without a plan, and the field placement is not ours to reschedule.
- **The Final Report and Review & Reflection, due Wednesday, December 9.** Grades are due to the university shortly after that and I cannot hold them open.

There is nothing to keep track of here. No tokens to count, no percentage per day, no ledger.

Privacy (FERPA)

Everything on the public course site is **de-identified** — no student names, work, or grades. In your own assignments, use **pseudonyms** and remove identifying details from any student work — in text, image, audio, or video — before it goes in.

Professional dispositions

As a candidate for teaching licensure, you are expected to meet professional standards of practice: demonstrating commitment and responsibility, building respectful professional relationships, and engaging in critical, reflective practice.

Student support

- **Howe Writing Center** — free, individualized writing help at any stage. (A draft may be taken here as the Review & Reflection read.) http://miamioh.edu/howe/
- **Rinella Learning Center** — academic counseling, tutoring, and supplemental instruction. https://miamioh.edu/student-life/rinella-learning-center/
- **English Language Learner Writing Center (ELLWC)** — support for multilingual writers. https://miamioh.edu/centers-institutes/english-language-learner-writing-center/

- **Student Counseling Service** — therapy, workshops, and wellness programming. https://miamioh.edu/life-at-miami/health-well-being/student-counseling-service/
- **Office of Student Wellness** — health and wellness education and resources. http://miamioh.edu/student-life/student-wellness/